

Expanding Universe

Year of Discovery: 1926

What Is It? The universe is expanding. The millions of galaxies move ever outward, away from its center.

Who Discovered It? Edwin Hubble

Why Is This One of the 100 Greatest?

Hubble's twin discoveries (that there are many galaxies in the universe—not just the Milky Way—and that all of those galaxies are traveling outward, expanding the universe) rank as the most important astronomical discoveries of the twentieth century. These discoveries radically changed science's view of the cosmos and of our place in it. Hubble's work also represents the first accurate assessment of the movement of stars and galaxies.

The discovery that the universe is expanding and ever changing for the first time allowed scientists to ponder the universe's past. This discovery led directly to the discovery of the Big Bang and the origin of the universe as well as to a new concept of time and of the future of the universe.

How Was It Discovered?

In 1923 Edwin Hubble was a tall, broad-shouldered, powerful astronomer of 33 who, 10 years earlier, had almost chosen a career as a professional boxer over astronomy. Hubble had been hired in 1920 to complete and operate the Mt. Wilson Observatory's mammoth 100-inch telescope in California—the largest telescope in the world.

In the early twentieth century, the universe was thought to contain one galaxy—the Milky Way—plus scattered stars and nebulae drifting around its edges. Hubble decided to use the giant 100-inch telescope to study several of these nebulae and picked Andromeda as his first target—and he made the two most important astronomical discoveries of the twentieth century.

This giant telescope's power showed Hubble that Andromeda wasn't a cloud of gas (as had been thought). It was a dense cluster of millions of separate stars! It looked more like a separate galaxy.

Then Hubble located several Cepheid stars in Andromeda. Cepheid stars pulse. The beat of their pulse is always a direct measure of the absolute amount of light given off by the star. By measuring their pulse rate and their *apparent* amount of light, scientists can determine the exact distance to the star.



Andromeda lay 900,000 light-years away. That proved that Andromeda was a separate galaxy. It lay too far away to be a fringe part of the Milky Way.

Within six months, Hubble had studied and measured 18 other nebulae. They were *all* separate galaxies, ranging from five to 100 million light-years from Earth. Astronomers were shocked to learn that the universe was so big and that it likely contained thousands of separate galaxies.

But Hubble was just beginning. He had noticed a consistent red shift when studying the light emitted from these distant nebulae.

Scientists had discovered that each element (helium, hydrogen, argon, oxygen, etc.) always emitted energy in a characteristic set of specific frequencies that identified the element's presence. If they made a spectrograph (a chart of the energy radiated at each separate frequency) of the light being emitted from a star, the lines on the spectrograph would tell them which elements were present in the star and in what relative quantities.

Hubble found all the common spectrograph lines for helium, hydrogen, and so forth that were normally found in a star. But all the lines on his graph were at slightly lower frequencies than normal. It was called a red shift because when visible light frequencies are lowered, their color shifts toward red. If their frequency is raised, their color shifts toward blue (a blue shift).

Over the next two years, Edwin Hubble conducted exhaustive tests of the 20 galaxies he had identified. He found that every one (except Andromeda) was moving *away* from Earth. More startling, the galaxies moved away from us *and* away from each other. Every galaxy he studied was speeding straight out into open space at speeds of between 800 and 50,000 kilometers per second!

The universe was expanding, growing larger every second as the galaxies raced outward. It was not a static thing that had remained unchanged since the beginning of time. In each moment the universe is different than it has ever been before.



Fun Facts: Because the universe is expanding, every galaxy in existence is moving away from our own Milky Way—except for one. Andromeda, our nearest neighbor, is moving on a collision course with the Milky Way. Don't worry, though: the collision won't occur for several million years.

More to Explore

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